

Machine Learning II Project

Evaluating *Using Machine Learning to Predict Realized Variance* by Peter Carr, Liuren Wu and Zhibai Zhang, in the Context of 2020 Market Volatility

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1 Introduction

The CBOE Volatility Index (VIX) [CBO20] is a measure of the expected 30-day future variance of the S&P 500 index (SPX) and is popularly used for tracking the SPX volatility. The index is calculated using a variance swap-based pricing formula using a weighted combination of out-of-the-money index options, whose prices are calculated in the risk neutral measure. This approach has multiple short-comings, including not being able to be bought and sold directly and the variance swap used to create the VIX is priced using a large number of options, including expensive and illiquid out of the money (OTM) options, making it very expensive to hedge.

Though the VIX cannot be bought or sold directly, a variety of methods have been proposed to replicate it [BD18], [CV19]. Rather than replicating the VIX directly, remembering that the true aim is to produce a successful estimator for the 30-day future variance of SPX, Carr et al. propose [Pet20] a suite of alternatives to the VIX. They use machine learning techniques to weight a selection of liquid options on SPX, to predict future variance.

Carr et al. present two main approaches. Firstly, a direct fit on 30-day volatility (referred to as Regression 2), and secondly creating a new VIX-like index, VIX^* , and subtracting VIX^{*2} from the 30-day future variance to calculate and then fit a VIX Risk Premium (VRP) (referred to as Regression 2), which Carr argues is a normalization procedure and therefore a more performant approach.

The VIX-alikes are all created and evaluated using a few common Machine Learning (ML) techniques, including Ridge Regression, a Feedforward Neural Network and a Random Forest. The advantages of all of these results are that by being a weighted portfolio of liquid options, the index should be cheap, tradable and may even hold more predictive power than the original VIX which is based on the outcome of a risk-neutral pricing.

In this paper we present our own VIX-like indices and VRP-like approach for predicting future variance, using the same approaches as Carr et al. We will pay special attention to the performance of these techniques during the period following March 2020 to evaluate the performance under the extreme stress of the onset of the Covid-19 pandemic and subsequent bull-run recovery in the S&P.

2 VIX and Variance Prediction

Carr et al. use transformed SPX option prices as features to predict 30-day SPX realized variances. The motivation behind the choice of features is that, VIX squared, often used to predict the 30-day future variance of SPX, is defined as a weighted sum of some transformed SPX option prices. Given the great cost to replicate VIX, it is natural to ask if machine learning algorithms can do a better job than a simple weighted sum in predicting the realized variances, while only using a smaller set of liquid options.

Specifically, VIX is defined in [CBO20] as

$$VIX = 100 \sqrt{T_1 \sigma_1^2 \left(\frac{T_2 - T_{30}}{T_2 - T_1} \right) + T_2 \sigma_2^2 \left(\frac{T_{30} - T_1}{T_2 - T_1} \right)}, \quad (1)$$

where

$$\sigma_i^2 = \frac{2}{T_i} \sum_h \frac{\Delta K_h}{K_h^2} e^{R_i T_i} O(K_h, T_i) - \frac{1}{T_i} \left(\frac{F_i}{K_0} - 1 \right)^2, \quad i = 1, 2$$

- F_i is the current T_i forward SPX index level
- K_0 is the first trading strike below F_i
- K_h 's are the strikes of other OTM strikes
- $\Delta K_h = K_{h+1} - K_h$ is the difference between two adjacent strikes
- $O(K_h, T)$ is the price of the T option (i.e. option that expires at T) with strike K_h
- T_{30} is 30 days from now
- T_1 and T_2 are the closest trading expiries around T_{30}
- and R_i is the (spot) interest rate at T_i

Eq. (1) states that $(VIX/100)^2$ is an interpolation between $\sigma_1^2 T_1$ and $\sigma_2^2 T_2$, which are the implied variances of the index at time T_1 and T_2 , respectively. This interpolation step is needed because the market has fixed maturities for options, so there will not necessarily be options with exactly T_{30} to expiry. On a date where there is, the VIX formula does not need an interpolation and reduces to

$$VIX = 100 \sqrt{T_{30} \sigma_{30}^2}, \quad \sigma_{30}^2 = \frac{2}{T_{30}} \sum_h \frac{\Delta K_h}{K_h^2} e^{R_{30} T_{30}} O(K_h, T_{30}), \quad (2)$$

where R_{30} is the spot interest rate at 1 month. Note that in this case the forward term $(F/K_0 - 1)^2/T_{30}$ vanishes as put call parity guarantees $F = K_0$. Eq. (2) can be rewritten as

$$(VIX/100)^2 = \sum_h 2 \Delta K_h e^{R_{30} T_{30}} \frac{O(K_h, T_{30})}{K_h^2}, \quad (3)$$

which is simply a weighted sum of $O(K_h, T_{30})/K_h^2$.

The left hand side expression of (3), $(VIX/100)^2$, is a popular estimator of the 30-day realized variance of SPX. When T_{30} options are not trading and the interpolation is applied, the formula of $(VIX/100)^2$ will be slightly more complicated, but it is still a weighted sum of the transformed option prices $O(K_h, T_1)/K_h^2$ and $O(K_h, T_2)/K_h^2$, plus two forward terms $-(F_i/K_0 - 1)^2/T_i$ that are usually tiny and can be ignored.¹ These transformed option prices are the features used by Carr et al. in their machine learning models.

The VIX* defined by Carr et al. is very similar to VIX. The only difference is the number of options considered. According to CBOE's definition of VIX, the number of options used for computation depends on the liquidity. SPX options have a fix strike difference of $\Delta K = 5$. For VIX computation, options at any strike K such that both $K + \Delta K$ and $K - \Delta K$ have option quotes are taken into the summation formula of σ_i^2 . If the option at a strike K is not quoted, and the option at one of $K + \Delta K$ and $K - \Delta K$ is not quoted either, this strike K is considered too deep OTM and the corresponding option price is not taken, and neither are any possibly trading but deeper OTM options. VIX*, however, simply fixes the number of options to be $2n + 1$ in the summation formula, that is n OTM call options and n OTM put options and the ATM option. Carr et al. experiment the performance of the prediction for various n .

¹At the time of this writing, the 1 month forward value F is close to 3900. Since SPX options have a fix strike difference of 5, we have $F - 5 < K_0 < F$. Thus the magnitude of the forward term is less than $12 \times ((3900 - 5)/3900 - 1)^2 = 0.00001972386$.

3 Data

Our data source is Bloomberg, and we have used it to retrieve the VIX and the following inputs to put into the Black-Scholes formula for generating option prices from January 2005 to February 2021. Specifically, we retrieve

- **SPX Index** to calculate the 30-day realized volatility of S&P 500 index and match with our estimate from option prices.
- **US001M Index** libor one month interest rate for Black-Scholes formula
- and finally implied volatilities (IVs) of options with 30-day (30D) expiry. We retrieve IVs for ATM, 10-delta, 25-delta and 40-delta puts and calls.

The queries to retrieve the data all follow the same flavour, using BQL

```
@BQL(
  "SPX Index",
  "dropna(implied_volatility(expiry=30D,delta = 25,put_Call = Call))",
  "dates=range(-20y,0d)",
  "cols=2;rows=4070"
)
```

3.1 Feature engineering

As any point on the implied volatility surfaces can be queried from Bloomberg, we simply retrieve the IVs of the T_{30} options, which we denote by σ_{30} . It should be noted that we are taking a shortcut here. CBOE's definition of VIX, and Carr's paper, perform the interpolation between the option prices at the closest trading expiry pairs around T_{30} . In reality this will not matter too much, and it really introduces a difference in feature as much as the interpolation approach Bloomberg takes on its implied volatility surface. A future extension to our implementation could be to create our IV series in a similar way, then our indices would be fully replicable.

The feature we intend on using is option price, which must be backed out of the x -delta option using Black-Scholes. The strike corresponding to the x -delta option can be backed out from the Black-Scholes formula for Delta. For example, the strike corresponding to the 10% delta call option can be obtained by solving

$$N\left(\frac{\log(S_0/K) + (R_{30} + \sigma_{30}^2/2)T_{30}}{\sigma_{30}\sqrt{T_{30}}}\right) = 10\%,$$

where S_0 is the current index level. Using this method, we were able to obtain time series of T_{30} option prices at out 7 strikes since 2005.

Following Carr et al. we apply a further transformation of the option prices $O(K_h, T_{30})/K_h^2$ as the features for our machine learning models. Each feature is finally normalized to zero mean and unit standard deviation.

We explore both target regressions discussed by Carr et al. and therefore must create two different target variables: y_{Reg1} , which is the 30-day (21 business day) realized variance of SPX retrieved from Bloomberg and secondly, the VIX Risk Premium (VRP) defined as

$$y_{Reg2} = VRP = VIX_t^{*2} - Var_t. \quad (4)$$

These features are shown through time here in figure 1.

4 Modelling

4.1 Calibration Technique

In the general context of machine learning we using a summarization function $f(\cdot)$ to find the best fit of a target y_{true} , based on some error function

$$\hat{f} = \operatorname{argmin} \{ \operatorname{Err}(f(X), y_{true}) \}$$

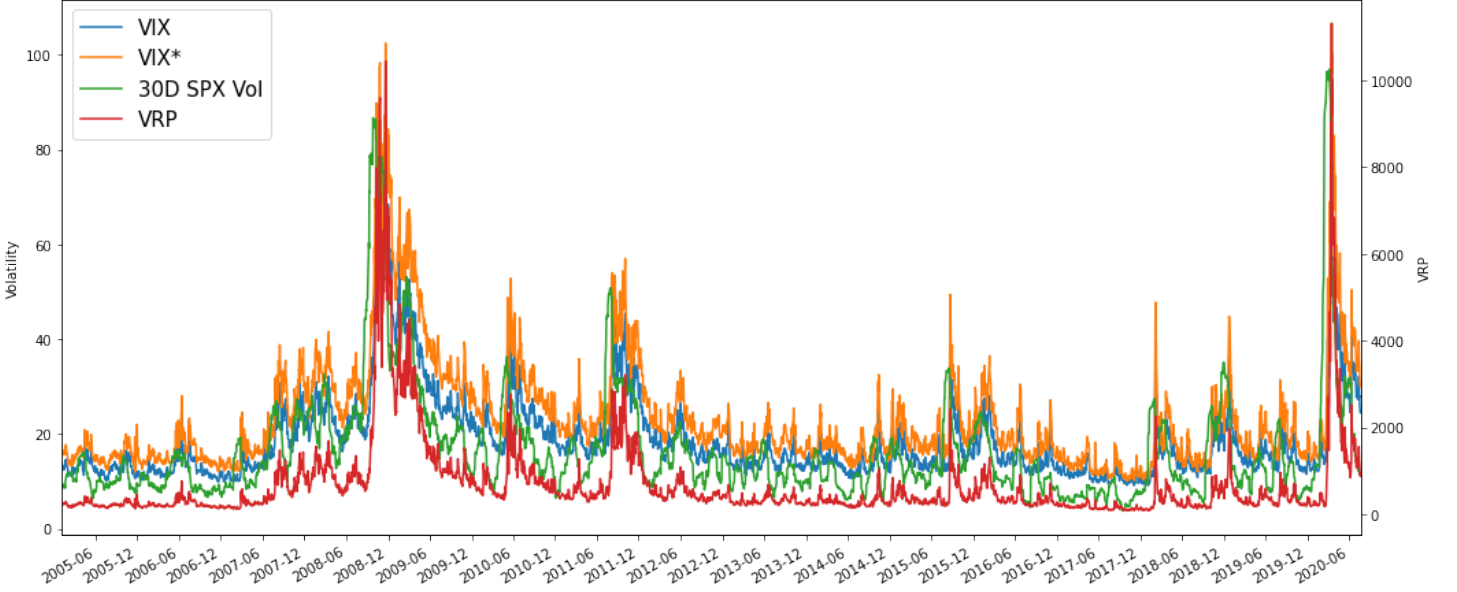


Figure 1: History of VIX, VIX* and actual SPX Historical Volatility and VRP

The result will be some new prediction

$$y_{\text{pred}} = \hat{f}(X)$$

and we'll be interested in the errors between y_{true} and y_{pred} .

For each potential model we will attempt to predict either the 30-day sample variance of return directly, y_{Reg1} , or VRP, y_{Reg2} . Our input features, X , will be the option prices $O_t(K_i, T_{30})$ as calculated in section 3.1. Our problem can be formulated as:

$$\text{Var}_t^T = f(\{O_t(K_i, T_j)\}) + \epsilon_t \quad (5)$$

$$f = \text{argmin} \{ \text{Err}(f(\{O_t(K_i, T_j)\}), \text{Var}_t^T) \} \quad (6)$$

4.1.1 Rolling windows

It will be necessary to judge how good a particular technique is at generalizing the relationship between X and y_{true} . To perform this on a time series, where it is logical to keep contiguous blocks of the time series together, it is not appropriate to split the dataset by randomly assigning elements into groups. Instead the data set is divided into k equally sized pieces and $k - 1$ model [training set, test set] iterations are created. Note that each training set gets successively larger, this is demonstrated in table 1.² Model performance can be evaluated on each test set.

	Period=1	Period=2	Period=3	Period=4	Period=5
Iteration=1	Train	Test			
Iteration=2	Train	Train	Test		
Iteration=3	Train	Train	Train	Test	
Iteration=4	Train	Train	Train	Train	Test

Table 1: Depiction of training and test splits for $k = 5$

²Unlike Carr et al. we opt for a simpler $k = 5$ fold, rather than a 30-day rolling window for convenience. In a production setting we could consider aligning resetting calibrations to option expiries, but as we are not using performing calibration based on expiry and we will not directly compare our performance to Carr et al. this seems sufficient

4.2 Algorithms

The main constraint behind the selection of a suitable ML algorithm is tradability. A trader should be able to construct a weighted portfolio the European SPX options to hedge the *VIX* index. The condition can be translated into the mathematical constraint that the prediction function should be piece-wise linear. Keeping this in mind, we select the same ML algorithms as in the paper.

4.2.1 Ordinary Least Squares

OLS though simple has an issue when forecasting volatility of daily returns that it is usually an overestimation in low volatility period and underestimation during high volatility period. In general, the volatility distribution has skew and is fat tailed. We find that the high volatility events are sparse among the low volatility periods and also volatility has the tendency to form clusters i.e. volatility remains high for number of days before reversing to normal.

4.2.2 Ridge Regression

Strong correlations between options will also pose issues of stability to OLS calibrations, so Ridge Regression techniques were also considered as they are known to handle correlated features gracefully. However, neither OLS or ridge are capable of handling non-linear features in the volatility distribution such as skew and clustering, so we expect ridge to have limited capacity to fit the dataset. The model was ran with cross validation for regularization strengths $\lambda \in [10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 10^2, 10^3, 10^4]$. On the latest training window, that which contained the most data λ from the CV process was chosen as 10^{-2} .

4.2.3 Random Forest

Random forests can also take care of strongly correlated features but this model will violate the tradability condition because it is not piece-wise linear. Despite being robust, it is also prone to overfitting as there are many parameters. However, we still applied the random forest for comparison. For the random forest we hand-tested many configurations including number of estimators $n \in [10, 50, 100]$ and max depth, $m \in [\infty, 3, 5, 10]$. Random Forests have a wealth of parameters which could be explored and a further study of suitable hyperparameter settings.

4.2.4 Neural Network

Finally, we use the Python package `pytorch` to build a FeedForward neural network (FNN) with the following functional form:

$$y(x, w) = O \left(\sum_j w_j^{(2)} \cdot h \left(\sum_i w_{ji}^{(1)} x_i + w_{j0}^{(1)} \right) + w_0^{(2)} \right) \quad (7)$$

where O is the activation function on the output layer and h is the activation function on the hidden layer. Here we select the rectified linear unit (ReLU) as the activation function to make y piecewise linear, so that the tradability constraint is satisfied. To see that y is piecewise linear, note that $\text{ReLU}(x) = \max\{0, x\}$ and hence

$$y(x, w) = \sum_{i,j} 1_{(2),j} w_j^{(2)} w_{ji}^{(1)} x_i + \text{const.} \quad (8)$$

The hidden layer has 7 nodes, same as the number of features. In figure 2 we see the results of our network calibration, a smooth fit over a 50 trial training period and successful mappings of features to the actual data.

4.3 Results

In this section we discuss our result for the two regression methods across our four ML algorithms. Along with it we also present feature importance plots in order to better understand the role of each options in the resultant model.

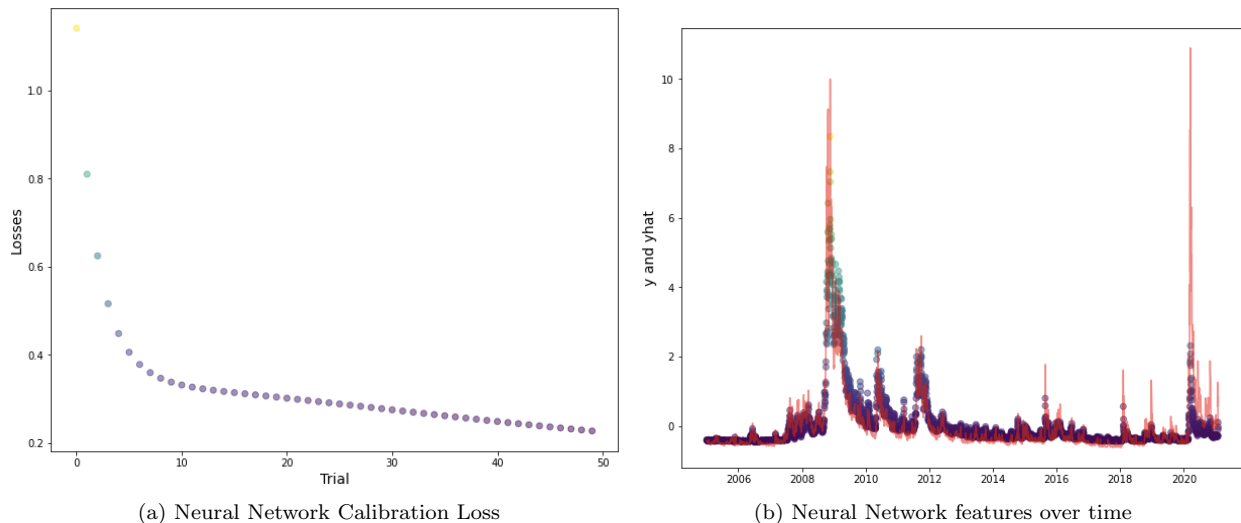


Figure 2: Training Results of the FeedForward Neural Network

4.3.1 Model Performance

For each time-sliced calibration period as discussed in section 4.1.1 we perform a forecast on next 30-days and collating all forecast windows together we generate one summary statistic, OOS R^2 which is calculated as one minus the ratio of the predicted variance to the actual variance.

$$R^2 = 1 - \frac{\sum_i (y_i - p_i)^2}{\sum_i (y_i - \bar{y})^2} \quad (9)$$

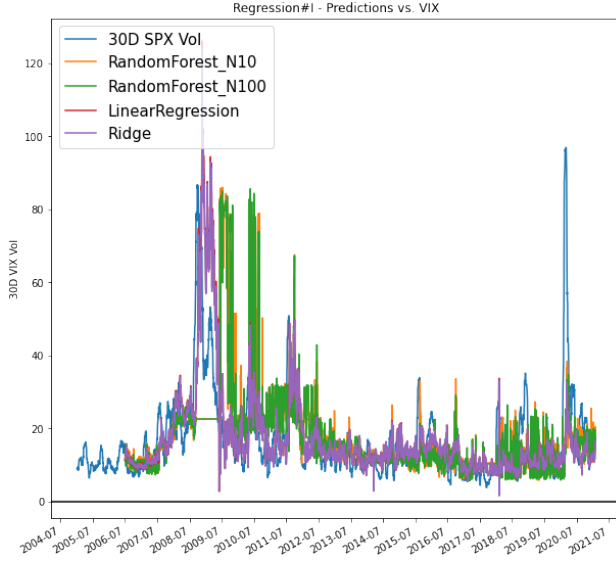
A summary of the results is provided in table 2

R^2	Regression I	Regression II
LinearRegression	-0.146	0.622
Ridge	-0.133	0.623
RandomForest_N10	-3.223	0.301
RandomForest_N100	-3.153	0.301
FFN_ReLU150	-	0.594

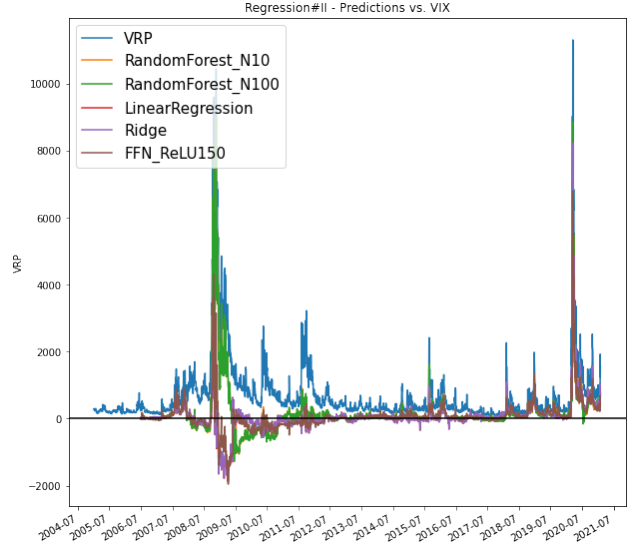
Table 2: OOS R^2

As in the paper, we find Regression II to have significantly better performance. Due to the small number of option strikes we were not able to replicate the findings of Carr et al. which asserted that as the number of strikes included increases the other model performances would increase and the linear regression would worsen. We have financial crisis 2008 and Coronavirus crisis 2020 in our option prices dataset making it difficult to fit the S&P realized variance in regression I. However fitting the VRP directly was much more convincing.

The most value for such a model is found during times of significant stress, the model would be most valuable if it were predictive of crashes. Here we show the actual vs. prediction values for our full dataset (figure 3) and specifically call out the two main crashes of the 21st century (thus far), the financial crisis of 2008 and the coronavirus pandemic from (figure 4), we see that the in general (and we see from our OOS R^2) that the Regression I fit is poor. However, Regression II, the fit for the VIX Risk Premium is very responsive to changes in VRP and the performance is a good fit for both the pandemic and financial crises periods.

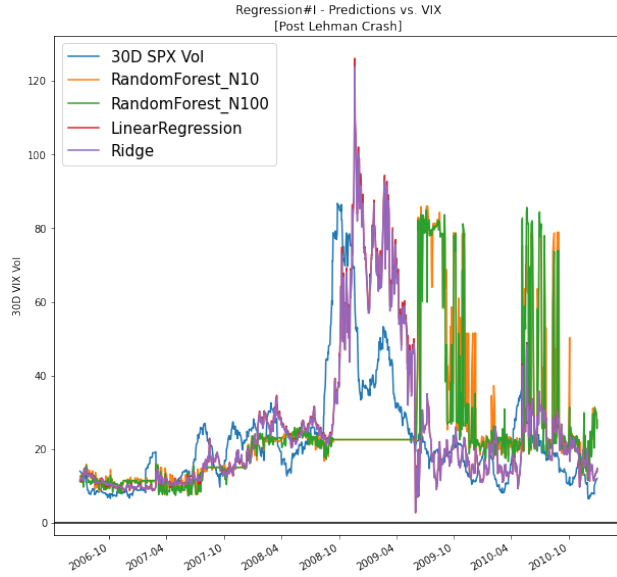


(a) Regression I

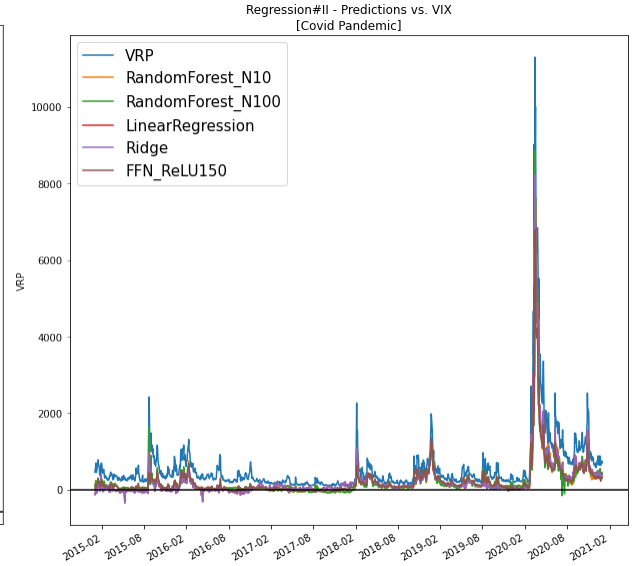


(b) Regression II

Figure 3: Full calculation period predictions vs. actuals



(a) Global Financial Crisis (Regression I)



(b) Coronavirus Period (Regression II)

Figure 4: Predict vs. actual historical volatilities

4.3.2 Feature Importance

In order to determine a particular feature's importance we cannot simply compare the weights between what VIX uses and what our regression methods generate. Larger weights do not necessarily correspond to higher importance in prediction. For this reason, we used a commonly used machine learning method called mean-decreased-accuracy (MDA) method.

The MDA procedure, as the paper by Carr et al. propose [Pet20] cites, computes the importance of each feature as follows: (a) train an algorithm on training set using all features; (b) predict on test set, and

compute performance (R^2 in our case) as p_0 ; (c) values of one feature i , in test set are randomly shuffled and predictions are made on test set; (d) performance for the shuffled feature i is recorded as p_i . Finally, MDA feature importance for i is computed as:

$$MDA(i) = (p_0 - p_i)/p_0 \quad (10)$$

We used Regression II with T=30days with 15 options using Ridge Regression. This is shown in Figure 5, we observed the same behaviour as in the paper by Carr et al. propose [Pet20], that: (a) for both put and calls, the more out-of-the-money higher the feature importance; (b) calls are on average more important than puts. This is different from VIX's weighting scheme. However in our FNN ReLU model we saw puts to be more important. (Figure 4). So we can't conclusively say that we agree with the paper. We would require more data to validate this.

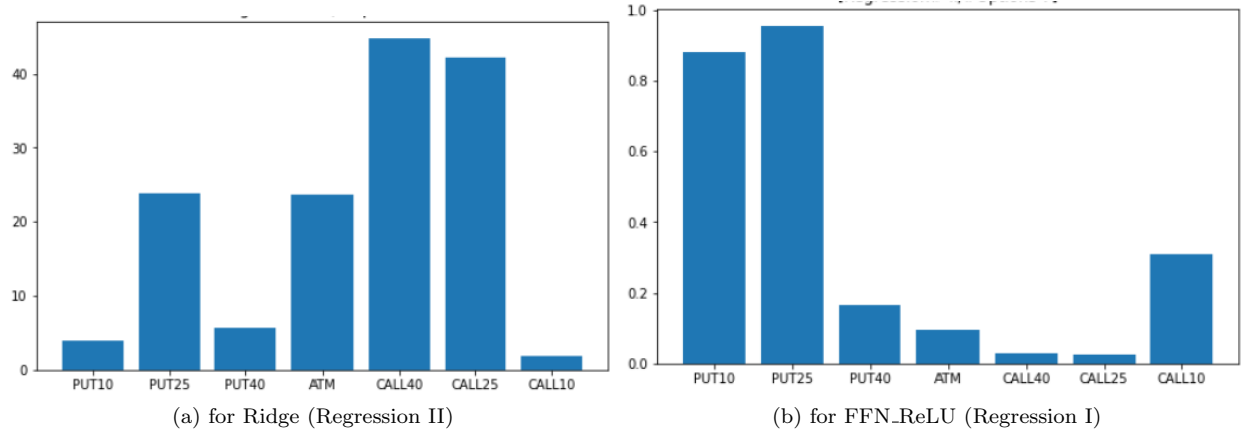


Figure 5: Feature Importance

5 Conclusion

Carr et al. aimed to formulate the problem of predicting the realized volatility of the SP 500 as a regression problem on option prices, using a suitable ML techniques. Their conclusion was that both Ridge regression and FNN have high predictive power. Further they suggested that the ML algorithm can predict the variance risk premium (VRP) i.e. difference between the realized volatility and the VIX better. Which we were able to replicate.

In our short study, instead we pulled the implied volatility data from the Bloomberg for a limited set of option delta for the period from 2005 to 2020 covering two big financial crisis, the Coronavirus crisis being the biggest one. Despite the limited set of option data, we could successfully fit linear, ridge, random forest and FNN ML algorithm on the option data set and reached to the same conclusion that fitting the variance risk premium (VRP) gives the better OOS R^2 using the same rolling OOS test procedure described in the paper.

We also conclude that the linear ridge and FNN with ReLU fit the best of the models that we explored. Besides FNN with ReLU meets the tradability condition. In their feature importance analysis, the authors found that the calls had higher importance and we reached to the same conclusion in Regression II for Ridge Regression. However this was not universal as we saw Puts having higher importance in FNN Relu Regression I.

Given the limited scope of this project, we did not repeat the analysis for 60 days tenors and more rich set of options. We also did not include other features such as historical volatility and returns such as in other ARCH/GARCH models. However we found our ML Algorithm is robust enough to handle the most recent Coronavirus crisis period in terms of its predictive power.

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